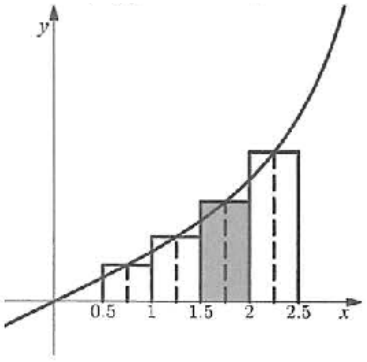
Numerical Methods

Questions 1–72

2.63 — Question 1. Part of the function $f(x) = \tan 0.4x$ is shown in the diagram below. Four rectangles of equal height have been sketched, each with height determined by $f(x)$ at their midpoints.



Find the area of the shaded rectangle, giving your answer to three significant figures.

2.63 — Question 2. Using the mid-ordinate rule, calculate an approximation to the integral

$$\int_0^2 e^{-x^2}\,\mathrm{d}x$$

when the width of the strip is given by each of the following values of h, giving your answer to six decimal places:

a) $h = 2$
b) $h = 1$
c) $h = 0.5$

2.63 — Question 3. Using the mid-ordinate rule, calculate an approximation to the integral

$$\int_1^2 \frac{1}{1-\ln x}\,\mathrm{d}x$$

for each of the following, giving your answer to six decimal places:

a) One strip.
b) Two strips.
c) Four strips.

2.63 — Question 4.

a) Use the mid-ordinate rule with four strips of equal width to find an estimate for

$$\int_0^2 \frac{\sin x}{x}\,\mathrm{d}x$$

giving your answer to three significant figures.

b) Explain how you could find an improved estimate.

2.63 — Question 5. The velocity of a car v m s^{-1} at time t seconds is modelled by

$$v = 10 + 3t - 0.2t^2 \quad \text{for } 0 \leq t \leq 10.$$

Use the mid-ordinate rule with five strips to estimate the distance travelled in the first 10 seconds.

2.63 — Question 6. Prerequisite knowledge for Question 6: Integration of $\ln x$.

a) Use the mid-ordinate rule with three rectangles to find an estimate for

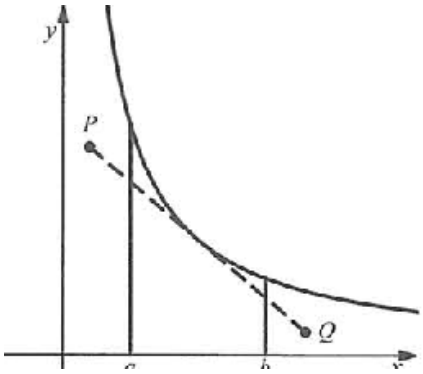
$$I = \int_1^4 \ln x \, \mathrm{d}x$$

giving your answer to four decimal places.

b) Find the exact value of I.

c) Find the relative error of your approximation from part (a).

2.63 — Question 7. The diagram shows part of the graph of $y = f(x)$. The line PQ lies on the tangent to the graph of $y = f(x)$ at the midpoint between a and b.



a) Use the mid-ordinate rule with one strip to find an expression to estimate the integral $\int_a^b f(x)\,\mathrm{d}x$ in terms of a and b.

b) State, giving reasons, whether you expect your estimate to be an under-estimate or an over-estimate.

2.63 — Question 8. The graph of $y = f(x)$ is concave (concave downwards) for the interval $[a, b]$. By considering a trapezium touching the curve at the midpoint of a and b, explain why the mid-ordinate rule with one strip overestimates $\int_a^b f(x)\,\mathrm{d}x$.

2.63 — Question 9.

a) Use the mid-ordinate rule with two strips of equal width to find an estimate for

$$\int_0^1 3\cos x \, \mathrm{d}x$$

giving your answer to two decimal places.

b) Using a sketch of the graph of $y = 3\cos x$ for $0 \leq x \leq 1$, explain clearly whether your answer to part (a) is an overestimate or an underestimate.

2.63 — Question 10. Prerequisite knowledge for Question 10: Hyperbolic Functions.

a) Use the mid-ordinate rule with four strips of equal width to find an estimate for

$$\int_1^2 \sinh x \, \mathrm{d}x$$

giving your answer to three significant figures.

b) Using a sketch of the graph of $y = \sinh x$ for $1 \leq x \leq 2$, explain clearly whether your answer to part (a) is an overestimate or an underestimate.

2.63 — Question 11. Prerequisite knowledge for Questions 11–13: Trapezium Rule and the Chain Rule. Let

$$I = \int_1^4 \sqrt{x^2 - 1}\,\mathrm{d}x.$$

a) Find an estimate for I using the trapezium rule with three strips, giving your answer to four decimal places.
b) Find an estimate for I using the mid-ordinate rule with three strips, giving your answer to four decimal places.
Given $f(x) = \sqrt{x^2 - 1}$ and additionally that $f'(x)$ is always decreasing for $1 \leq x \leq 4$,
c) State, giving a reason, whether the trapezium rule gives an over- or under-estimate.
d) State, giving a reason, whether the mid-ordinate rule gives an over- or under-estimate.

2.63 — Question 12. Let

$$I = \int_2^{10} \frac{1}{\ln x}\,\mathrm{d}x.$$

a) Find an estimate for I using the trapezium rule with four strips, giving your answer to four decimal places.
b) Find an estimate for I using the mid-ordinate rule with four strips, giving your answer to four decimal places.
It is given that $f(x) = \dfrac{1}{\ln x}$.
c) Show that $f'(x)$ is always increasing for $2 \leq x \leq 10$.
d) State, giving a reason, whether the trapezium rule gives an over- or under-estimate.
e) State, giving a reason, whether the mid-ordinate rule gives an over- or under-estimate.

2.63 — Question 13. Let

$$I = \int_1^3 \frac{e^x}{x}\,\mathrm{d}x.$$

a) Find an estimate for I using the trapezium rule with two strips, giving your answer to five decimal places.

b) Find an estimate for I using the mid-ordinate rule with two strips, giving your answer to five decimal places.

It is known that a better approximation for I is 8.03871 to five decimal places.

c) What does this tell you about the concavity of the curve for $1 \leq x \leq 3$?

2.63 — Question 14. Prerequisite knowledge for Question 14: Volume of Revolution. A curve has equation $y = \ln(x+1)$. The region bounded by the curve, the lines $x = 0$ and $x = 2$, and the x-axis is rotated 360° about the x-axis.

a) Write down an expression for the exact volume of revolution.

b) Use the mid-ordinate rule with four strips to estimate this volume, giving your answer to three significant figures.

2.63 — Question 15. A curve has equation $y = \sin(x^2)$.

a) Use the mid-ordinate rule with two strips to find an estimate for

$$\int_0^1 \sin(x^2)\,\mathrm{d}x$$

giving your answer to four decimal places.

The graph with equation $y = \sin(x^2)$ is stretched with scale factor 3 parallel to the y-axis and then translated by $\begin{pmatrix}2\\0\end{pmatrix}$ to give the graph with equation $y = f(x)$.

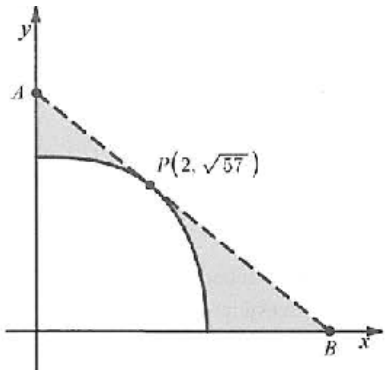
b) Write down an expression for $f(x)$.

Giving your answer to three decimal places:

c) Using your answer to part (a), state an estimate to $\int_2^3 f(x)\,\mathrm{d}x$.

d) Use the mid-ordinate rule with two strips to find an estimate for $\int_2^3 f(x)\,\mathrm{d}x$.

2.63 — Question 16. The shaded regions on the diagram below are bounded by the curve with equation $y = \sqrt{81 - 3x^3}$, $x \geq 0$, the tangent at P and the coordinate axes. The point P on the curve has coordinates $(2, \sqrt{57})$.



a) Use the mid-ordinate rule with three rectangles to find an estimate for

$$\int_0^3 \sqrt{81 - 3x^3}\,\mathrm{d}x$$

giving your answer to two decimal places.

b) Find the exact form of the equation of the tangent line, giving your answer in the form $ay + b\sqrt{57}x = c\sqrt{57}$, where a, b and c are integers.

c) Hence find an approximate value for the total area of the shaded regions, giving your answer to three significant figures.

2.63 — Question 17. Prerequisite knowledge for Question 17: Mean Value.

a) Sketch the curve C with equation $y = e^{x/2}$ for $0 \leq x \leq 3$.

The region bounded by the curve C, the coordinate axes and the vertical straight line with equation $x = 3$ is to be estimated by the mid-ordinate rule using three equally spaced strips.

b) Find, correct to six decimal places, the area of this region.

c) Hence estimate the mean value of $y = e^{x/2}$ in the interval $0 \leq x \leq 3$, giving your answer to five decimal places.

2.63 — Question 18. Prerequisite knowledge for Questions 18–19: Maclaurin Series.

a) State the first three non-zero terms in the Maclaurin expansion of $\ln(1 + x)$.

b) Use the mid-ordinate rule with four strips and this expansion to estimate

$$\int_{0.1}^{0.5} \ln(1 + x)\,\mathrm{d}x$$

giving your answer as a simplified fraction.

2.63 — Question 19.

a) Find the first three non-zero terms in the Maclaurin expansion of xe^x.

b) Use the mid-ordinate rule with six strips and this expansion to estimate

$$\int_0^{0.3} xe^x \, \mathrm{d}x$$

giving your answer to four decimal places.

2.64 — Question 1. Using Simpson's rule, calculate an approximation to the integral

$$\int_0^2 e^{-x^3} \, dx$$

for each of the following, giving your answer to 6 decimal places:

a) Three ordinates.

b) Five ordinates.

c) Seven ordinates.

2.64 — Question 2. Using Simpson's rule, calculate an approximation to the integral

$$\int_1^2 \frac{1}{1+\ln x}\,dx$$

for each of the following, giving your answer to 6 decimal places:

a) Two strips.
b) Four strips.
c) Six strips.

2.64 — Question 3. The table below shows the velocity $v\,\mathrm{m\,s^{-1}}$ of a car every 2 seconds.

t	0	2	4	6	8
v	0	5.6	9.8	13.0	15.2

Use Simpson's rule to estimate the distance travelled in the first 8 seconds.

Prerequisite knowledge for Q4: Hyperbolic Functions

2.64 — Question 4. Use Simpson's rule with five ordinates to find an estimate for

$$\int_0^1 \cosh x \, dx,$$

giving your answer to three significant figures.

2.64 — Question 5.

a) Use Simpson's rule with four strips of equal width to find an estimate for

$$\int_{0.1}^{0.5} \frac{\tan x}{x} \, dx,$$

giving your answer to 3 significant figures.

b) Explain how you could find an improved estimate.

2.64 — Question 6.

a) Use Simpson's rule with three ordinates to estimate

$$\int_0^{\pi} e^{\cos x}\, dx,$$

giving your answer to four decimal places.

b) Explain how Simpson's rule could be used to find a more accurate estimate of the integral in part (a).

2.64 — Question 7.

a) Use Simpson's rule with four strips to find an estimate for

$$\int_0^{1} e^{x}\, dx,$$

giving your answer to five significant figures.

b) Find the exact value of $\int_0^1 e^x\, dx$.

c) Find the relative error of your approximation from part (a).

Prerequisite knowledge for Q8: Trapezium Rule

2.64 — Question 8. It is given that

$$I = \int_0^1 (x^2 + e^{2x})\, dx.$$

a) Find an estimate for I using the trapezium rule with four strips, giving your answer to four decimal places.
b) Find an estimate for I using Simpson's rule with four strips, giving your answer to four decimal places.
c) Find the exact value of I.

Prerequisite knowledge for Q9: Reversing the Chain Rule or Integration by Substitution

2.64 — Question 9.

a) Use Simpson's rule with two strips to find an estimate for

$$\int_0^2 x\sqrt{2x^2+1}\, dx,$$

giving your answer to four decimal places.
b) Find the exact value of $\displaystyle\int_0^2 x\sqrt{2x^2+1}\, dx$.
c) Find the relative error of your approximation from part (a).

2.64 — Question 10. A curve has equation $y = \cos(x^2)$.

a) Use Simpson's rule with four strips to find an estimate for $\int_0^1 \cos(x^2)\,dx$, giving your answer to four decimal places.

b) The graph with equation $y = \cos(x^2)$ is stretched with scale factor 5 parallel to the y-axis and then translated by 1 parallel to the x-axis to give the graph with equation $y = f(x)$. Write down an expression for $f(x)$.

c) Giving your answers to three decimal places, using your answer to part (a), state an estimate for:

i) $\int_1^2 f(x)\,dx$.

d) Use Simpson's rule with four strips to find an estimate for $\int_1^2 f(x)\,dx$.

2.64 — Question 11.

a) Use Simpson's rule with five ordinates to find an approximation to

$$\int_{0.1}^{0.3} \frac{1}{\sqrt{1-x^3}}\,dx,$$

giving your answer to three decimal places.

b) Hence find an approximation to $\int_{0.1}^{0.3} \frac{5}{\sqrt{1-x^3}}\,dx$.

2.64 — Question 12.

a) Use Simpson's rule with five ordinates to find an approximation to

$$\int_1^3 \frac{\sin x}{x}\, dx,$$

giving your answer to three decimal places.

b) Hence find an approximation to $\int_1^3 \left(3 - \frac{\sin x}{x}\right) dx$.

Prerequisite knowledge for Q13: Volume of Revolution

2.64 — Question 13. A curve has equation $y = x\ln(x+2)$. The region bounded by the curve, the lines $x = 0$ and $x = 2$, and the x-axis is rotated 360° about the x-axis.

a) Write down an expression for the exact volume of revolution.

b) Use Simpson's rule with five ordinates to estimate this volume, giving your answer to 3 significant figures.

Prerequisite knowledge for Q14: Mean Value

2.64 — Question 14.

a) Sketch the curve C with equation $y = 1 + \sin(\frac{x}{2})$ for $0 \leq x \leq 2\pi$.

b) The region bounded by the curve C, the coordinate axes, and the vertical straight line with equation $x = 2\pi$ is to be estimated by Simpson's rule using six equally spaced strips. Find, correct to six decimal places, an approximation to the area of this region.

c) Hence estimate the mean value of $y = 1 + \sin(\frac{x}{2})$ in the interval $0 \leq x \leq 2\pi$, giving your answer to five decimal places.

2.64 — Question 15. Given that $\ln a = \displaystyle\int_1^a \frac{1}{x}\,dx$, use Simpson's rule with seven ordinates to find an estimate for $\ln 7$, giving your answer to four decimal places.

Prerequisite knowledge for Q16–Q17: Maclaurin Series

2.64 — Question 16.

a) Find the first three non-zero terms in the Maclaurin expansion of $\sin 2x$.

b) Use Simpson's rule with four strips and this expansion to estimate $\int_0^1 \sin 2x \, dx$, giving your answer to four decimal places.

2.64 — Question 17.

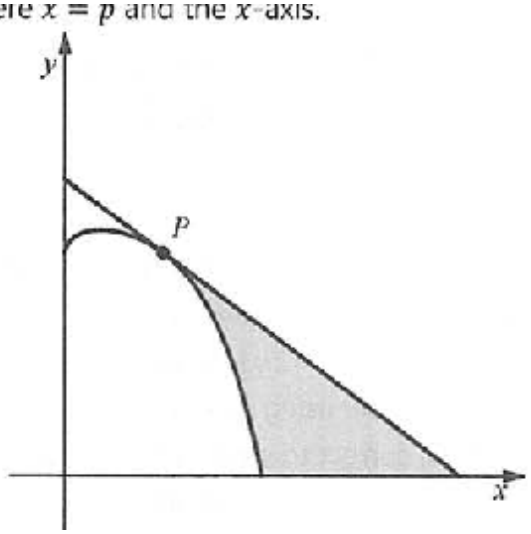
a) Find the first three non-zero terms in the expansion of $x^2 \ln(1+x)$.

b) Use Simpson's rule with four strips and this expansion to estimate $\int_0^{0.4} x^2 \ln(1+x) \, dx$, giving your answer to four decimal places.

Prerequisite knowledge for Q18: The Chain Rule

2.64 — Question 18. The shaded regions on the diagram are bounded by the curve with equation $y = -x^x + 4$ for $1 \leq x \leq 2$, the tangent line at point P where $x = p$, and the x-axis. It is given that $p = 1$.

where $x = p$ and the x-axis.



a) State the coordinates of the point P.

b) Use Simpson's rule with four strips to find an estimate for $\int_1^2 (-x^x + 4)\,dx$, giving your answer to two decimal places.

c) Given that $x^x = e^{x \ln x}$, show that $\dfrac{d}{dx}(x^x) = x^x(1 + \ln x)$.

d) Find the exact form of the equation of the tangent line, giving your answer in the form $ax + by = c$, where a, b and c are integers.

e) Hence find an approximate value for the total area of the shaded regions, giving your answer to two decimal places.

Prerequisite knowledge for Q19–Q21: Trapezium Rule, Mid-ordinate Rule and Reversing the Chain

2.64 — Question 19. Let $I = \int_0^2 x^2\,dx$.

a) Find an estimate for I, giving your answer as accurately as possible, using:
   i) the trapezium rule with four strips, T_4;
   ii) the mid-ordinate rule with four strips, M_4;
   iii) Simpson's rule with eight strips, S_8.

b) Find the exact value of I.

c) Show that $\dfrac{2M_4 + T_4}{3} = S_8$.

2.64 — Question 20. Let $I = \int_0^6 \sqrt{x+1}\,dx$.

a) Find an estimate for I, giving your answer to six decimal places, using:
   i) the trapezium rule with four strips, T_4;
   ii) the mid-ordinate rule with four strips, M_4;
   iii) Simpson's rule with eight strips, S_8.

b) Find the exact value of I.

c) Show that $\dfrac{2M_4 + T_4}{3} = S_8$.

2.64 — Question 21. Let $I = \int_a^b x^2\,dx$, where a and b are positive constants such that $a < b$. Prove each of the following:

a) The trapezium rule with two strips, T_2, is given by $\dfrac{1}{8}(b-a)(3a^2 + 2ab + 3b^2)$.

b) The mid-ordinate rule with two strips, M_2, is given by $\dfrac{1}{16}(b-a)(5a^2 + 6ab + 5b^2)$.

c) Simpson's rule with four strips, S_4, is given by $\dfrac{1}{3}(b-a)(a^2 + ab + b^2)$.

d) The exact value of I is given by $\dfrac{1}{3}(b-a)(a^2 + ab + b^2)$.

e) $\dfrac{2M_2 + T_2}{3} = S_4$.

2.65 — Question 1. The function $y(x)$ satisfies $\frac{dy}{dx} = e^{-x^2}$ where $y(2) = 1$. For each of the following, use $y_{n+1} = y_n + hf(x_n)$ to approximate $y(3)$ with the given value of h, giving your answer to three decimal places:

a) $h = 0.5$

b) $h = 0.25$

2.65 — Question 2. The function $y(x)$ satisfies $\frac{dy}{dx} = x^2 - y$ where $y(0) = 1$. For each of the following, use $y_{n+1} = y_n + hf(x_n, y_n)$ to approximate $y(0.4)$ with the given value of h, giving your answer to three decimal places:

a) $h = 0.2$

b) $h = 0.1$

2.65 — Question 3. The function $y(x)$ satisfies $\frac{dy}{dx} = \sin x - y$ where $y(0) = 2$. Calculate $y(0.4)$ using Euler's step by step method for the given value of h, giving your answer to three decimal places:

a) $h = 0.2$

b) $h = 0.1$

2.65 — Question 4. The function $y(x)$ satisfies the differential equation $\frac{dy}{dx} = \frac{x^2 + y}{y^2 - x}$ and the condition $y(2) = -1$. Use two iterations of Euler's step by step method to estimate the value of $y(3)$.

2.65 — Question 5. For each of the following, use Euler's step by step method:

a) $2x\dfrac{dy}{dx} = e^{\sqrt{x}}$ passes through the point $(4, 1)$. Estimate $y(5)$ using $h = 0.5$, giving your answer to three decimal places.

b) $\dfrac{dy}{dx} = \ln(x^2 + y)$ passes through the point $(0, 2)$. Estimate $y(1.5)$ using $h = 0.5$, giving your answer to three decimal places.

2.65 — Question 6. The function $y(x)$ satisfies $\dfrac{dy}{dx} = 2e^{2x}$ and the condition $y(0) = 1$.

a) Use Euler's step by step method with $h = 0.2$ to approximate $y(0.6)$, giving your answer to three decimal places.

b) Find the exact value of $y(0.6)$.

c) Find the relative error.

d) State whether the Euler approximation to $y(0.6)$ is an over- or an under-estimate of the exact value, giving a reason for your answer.

2.65 — Question 7. The function $y(x)$ satisfies $\frac{dy}{dx} = x + 3\cos y$ and the condition $y(1) = 4$.

a) Use Euler's step by step method with $h = 0.4$ to approximate $y(2.2)$, giving your answer to three decimal places.
b) Comment on the reliability of your result.
c) Explain how you could use this formula to find an improved approximation to $y(2.2)$.

2.65 — Question 8. A drug concentration C mg depends on how much is already in the bloodstream and the time t hours since infusion started. It may be modelled by $\frac{dC}{dt} = 5 - 0.2C + 0.1t$. Initially there is no drug in the bloodstream. Use Euler's method with $h = 0.5$ hours to estimate the concentration after 1 hour.

2.65 — Question 9. A town's population y of people grows depending on the current population and the season. It may be modelled by $\frac{dy}{dx} = 0.005y + 200\sin\left(\frac{\pi}{6}x\right)$ where x is the number of months passed and is measured in radians. Initially the population is 12,000 people. Use Euler's method with $h = 1$ month to approximate the population after 3 months.

2.65 — Question 10. The temperature of a rod $T°$C over time t minutes may be modelled by $\frac{dT}{dt} = 0.2t - 0.05T$. When 5 minutes have passed, the rod is at 50°C. Use Euler's method with $h = 1$ minute to estimate the temperature after 8 minutes, giving your answer to three significant figures.

2.65 — Question 11. The balance £B on an account grows depending on both balance and a time-varying interest rate, and may be modelled by $\frac{dB}{dt} = 0.01B + 5\cos(0.5t)$ where t months have passed. The account is checked after 2 months and the balance is £970. Use Euler's method twice to estimate the balance after a further 2 months.
Prerequisite Knowledge for Q12: Integration of Standard Functions

2.65 — Question 12. The function $y(x)$ passes through $(0, 1)$ and satisfies the differential equation $\frac{dy}{dx} = \sin x$.

a) Use three iterations of Euler's step by step method to estimate $y(0.3)$, giving your answer to three decimal places.
b) Explain why this method underestimates the true value of $y(0.3)$.

2.65 — Question 13. A curve satisfies $\frac{dy}{dx} = \frac{2x - y}{xy}$ and passes through the point $(3, a)$, where $a > 2$. Using Euler's step by step method, with starting point $(3, a)$, gives an estimate of $y = 4.450$ when $x = 3.2$. Find the value of a.

2.65 — Question 14. The function $y(x)$ satisfies the differential equation $\frac{dy}{dx} = \sqrt{2x + y^2}$ and the condition $y(1) = 0.8$. After one iteration of Euler's formula, $y_{n+1} = y_n + \frac{1}{k} f(x_n)$, for integer k, it is found that $y_1 = 1.1250$, which was correct to four decimal places.

a) Find the value of $\frac{1}{k}$.

b) Use your value for $\frac{1}{k}$ to calculate (x_2, y_2).

Prerequisite Knowledge for Q15: First Order Linear Differential Equations

2.65 — Question 15. The curve $y(x)$ satisfies $\dfrac{dy}{dx} = x + y$ where $y(0) = 1$.

a) Explain geometrically what the Euler formula $y_{n+1} = y_n + h(x_n + y_n)$ represents.

b) Use Euler's method with step size $h = 0.5$ to approximate $y(1)$.

c) Find the exact value of $y(1)$.

d) On a sketch, show the curve for $x > 0$, the tangent at $(0, 1)$ and the first Euler step.

2.65 — Question 16. A company monitors the total amount of data y GB uploaded to its servers during a live event. It is observed that the upload rate is directly proportional to the time t hours since the event began. After 4 hours, the upload rate is measured to be 24 GB per hour.

a) Determine the differential equation to model the total data uploaded.

It is observed that one hour into the event, 1.5 GB has been uploaded.

b) Use Euler's method with $h = 0.25$ to find the amount of data processed after 90 minutes, giving your answer to two decimal places.

2.65 — Question 17. A post on a social media platform spreads as users share it. At time t hours since posting, the number of people who have seen the post is N. The rate of change of N is affected by:

- sharing at a rate proportional to the current reach, $0.08N$ people per hour
- a reduction in visibility due to platform algorithms at a rate of $500e^{-t}$ people per hour

After 1 hour, the number of people who have seen the post is 4700. Use Euler's method with a step length of 0.5 to estimate the number of people who have seen the post after 2 hours have passed.

2.65 — Question 18. A leaking container is being monitored. The volume of water W litres changes over time t hours due to three factors:

- Rainfall adds water at an average constant rate of 0.5 litres per hour.
- Evaporation over each day removes water at a rate of $2\sin^2\left(\frac{\pi}{12}t\right)$ litres per hour.
- Leakage removes water proportional to the current volume in the pond, at a rate of $0.01W$ litres per hour.

At the start of observation (at time $t = 0$), the pond contains 100 litres of water. Use Euler's step by step method three times to estimate the amount of water in the container after 9 hours have passed. Give your answer to three significant figures.

2.66 — Question 1. The function $y(x)$ satisfies

$$\frac{dy}{dx} = (x-1)\ln(2x+y), \qquad y(1) = 2$$

a) Use the Euler formula $y_{r+1} = y_r + h\mathrm{f}(x_r, y_r)$ with $h = 0.5$ to obtain an approximation to $y(1.5)$
b) Using your answer to part a), use the formula $y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r)$ to obtain $y(2)$ giving your answer to three decimal places.

2.66 — Question 2. The function $y(x)$ satisfies

$$\frac{dy}{dx} = x + \ln(1+y), \qquad y(2) = 1$$

Use Euler's step by step method once, followed by using the midpoint formula

$$y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r),$$

where $x_{r+1} = x_r + h$ with $h = 0.1$ to obtain an approximation to $y(2.2)$ giving your answer to three decimal places.

2.66 — Question 3. Given a second point has already been found using Euler's step by step method, use the midpoint formula

$$y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r),$$

where $x_{r+1} = x_r + h$ for each of the following, giving your answers to three decimal places:

a) $\frac{dy}{dx} = y \times 2^x, (0, 1), (0.75, 1.75)$ and $h = 0.75$ to find $y(1.5)$

b) $\frac{dy}{dx} = y \tan x, (1, 0.2), (1.25, 0.2779)$ and $h = 0.25$ to find $y(1.75)$

2.66 — Question 4. Use the improved Euler's formula

$$y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$$

where $k_1 = h\mathrm{f}(x_r, y_r), k_2 = h\mathrm{f}(x_r + h, y_r + k_1)$ for each of the following. In each case the curve satisfies the given differential equation and passes through the given point. Give your answer to three significant figures:

a) $\frac{dy}{dx} = \ln(2x + \sqrt{y}), (1, 3)$ with $h = 2$ to find $y(3)$

b) $\frac{dy}{dx} = x - \cos^2 y, (2, 0.5)$ with $h = 0.2$ to find $y(2.2)$

c) $\frac{dy}{dx} = (3x + 2)\sqrt{x + y}, y(0) = 3$ with $h = 0.5$ to find $y(0.5)$

Prerequisite Knowledge for Q5: Separation of Variables

2.66 — Question 5. A curve passes through the point $(4, e^{0.1})$ and satisfies the differential equation

$$\frac{dy}{dx} = y\sqrt{2x+1}$$

for $x > -\frac{1}{2}$

a) Use Euler's step by step method once, then the midpoint formula

$$y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r), \quad x_{r+1} = x_r + h$$

once, each with a step length of 0.1 to estimate the value of y when $x = 4.2$
Give your answer to three decimal places.

b) Find the particular solution of the differential equation $\frac{dy}{dx} = y\sqrt{2x+1}$ in the form $\ln y = \mathrm{g}(x)$ where $\mathrm{g}(x)$ is function to be found.

c) Find the relative error of your approximation in part a) giving your answer to two significant figures.

Prerequisite Knowledge for Q6: First Order Linear Differential Equations

2.66 — Question 6. A curve passes through the point $(0, 1)$ and satisfies the differential equation

$$\frac{dy}{dx} = x - \frac{1}{2}y$$

a) Use Euler's step by step method once, then the midpoint formula

$$y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r), \quad x_{r+1} = x_r + h$$

once, each with a step length of 0.05 to estimate the value of y when $x = 0.1$

b) Find the general solution of the differential equation $\frac{dy}{dx} = x - \frac{1}{2}y$

c) Given that y satisfies the differential equation in part b) and that $(0, 1)$ lies on the curve, find the value of y when $x = 0.1$ giving your answer to three significant figures.

d) Comment on your accuracy in part a)

2.66 — Question 7. A patient's drug concentration C mg/L in the bloodstream changes over time (t hours) according to both metabolism and a slow continuous infusion. The rate of change of concentration satisfies

$$\frac{dC}{dt} + 0.2C = 1.5e^{-0.1t}$$

Initially, the patient's drug concentration is 0 mg/L

a) Use Euler's method, with a step size of 0.5, to find an approximation for the drug concentration after 30 minutes.
b) Explain how the numerical method used in part a) could be made more accurate.
c) Refine the approximation found in part a) using $y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$ where $k_1 = h\mathrm{f}(x_r, y_r)$ and $k_2 = h\mathrm{f}(x_r + h, y_r + k_1)$ with a step size of 0.5. Give your answer to two significant figures.

2.66 — Question 8. The temperature $T(°\mathrm{C})$ of a chemical solution at time t minutes is modelled by

$$(15 - 0.04t^2)\frac{dT}{dt} + 0.8T = 6t$$

The initial temperature of the solution is 20°C.

a) Use Euler's method, with a step size of 0.5, to find an approximation for the temperature of the solution after 1 minute, giving your answer to three significant figures.
b) Explain how the numerical method used in part a) could be made more accurate.
c) Refine the approximation found in part a) using the formula

$$y_{n+1} = y_n + h\mathrm{f}\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}\mathrm{f}(x_n, y_n)\right)$$

with a step size of 1. Give your answer to three significant figures.

2.66 — Question 9. The function $y(x)$ satisfies

$$\frac{dy}{dx} = x - y$$

with the initial condition $y(0) = 1$. The fourth-order Runge-Kutta method is defined by

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where $k_1 = \mathrm{f}(x_n, y_n)$

$$k_2 = \mathrm{f}\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$$

$$k_3 = \mathrm{f}\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right)$$

and $k_4 = \mathrm{f}(x_n + h, y_n + hk_3)$
Use the fourth-order Runge-Kutta method with step length $h = 0.5$ to obtain an approximation to $y(0.5)$. Give your answer to three decimal places.

2.66 — Question 10. A curve passing through the point $(1, a)$ where a is a constant satisfies

$$\frac{dy}{dx} = 5x - y$$

Applying Euler's step by step method once using $h = 0.1$, followed by the formula $y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r)$ where $x_{r+1} = x_r + h$ once from starting point $(1, a)$ gives an estimate of $y = 3.05$ when $x = 1.2$
Find the value of a

2.66 — Question 11. The function $y(x)$ satisfies

$$\frac{dy}{dx} = x^2 - y$$

and the condition $y(1) = 0.5$. After applying Euler's step by step method once, it is found that $y_1 = 0.6$

a) Find the value of h giving your answer to one decimal place.

b) Use the improved Euler's method, where $y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r)$ and $x_{r+1} = x_r + h$, to find (x_2, y_2)

2.66 — Question 12. A curve passes through $(0, 1)$ and satisfies

$$\frac{dy}{dx} = x + y$$

Applying Euler's step by step method once, followed by $y_{r+1} = y_{r-1} + 2h\mathrm{f}(x_r, y_r)$ where $x_{r+1} = x_r + h$ once it is found that $y_2 = 1.96$

Determine the value of h

2.66 — Question 13. A curve which passes through the point $(2, a)$ where a is a constant, satisfies

$$\frac{dy}{dx} = 2y - 3x$$

Using $y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$ where $h = 0.5$, $k_1 = h\mathrm{f}(x_r, y_r)$ and $k_2 = h\mathrm{f}(x_r + h, y_r + k_1)$ with starting point $(2, a)$ gives an estimate of $y = 7.625$ when $x = 2.5$. Determine the value of a

2.66 — Question 14. A curve passes through $(0, 1)$ and satisfies

$$\frac{dy}{dx} = 4y + 2x$$

The formula $y_{r+1} = y_r + \frac{1}{2}(k_1 + k_2)$ where $k_1 = h\mathrm{f}(x_r, y_r)$ and $k_2 = h\mathrm{f}(x_r + h, y_r + k_1)$ is used to find that $y_1 = 2.16$ exactly. Determine the value of h